

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	23.50
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	26.80
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	26.29
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	26.29

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$M(\text{BaCl}_2) = 137.3 + (35.45 \times 2) = 208.2 \text{ g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} \quad n(\text{BaCl}_2) = \frac{137.3 \times 10^{-23}}{208.2 \times 10^{-3}} = 1.032 \text{ mol}$$

2.00 mol

1 for units.

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$0.51 \text{ g of water lost.}$$

- b) Calculate the moles of water lost in the heating process

$$n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{0.51 \times 10^{-23}}{18.016 \times 10^{-3}} = 0.283 \text{ mol}$$

0.3 mol

3

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

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Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

It means that the measurements had varying amounts of water combined with them.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

The desiccator was used to ensure that no water combined with the BaCl_2 to increase its mass.

from the air

TOTAL = 7.5 / 15

50%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	22.24g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	26.01g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	25.42g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	25.42g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$25.42\text{g} - 22.24\text{g} = 3.18\text{g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} \quad n(\text{BaCl}_2) = \frac{3.18\text{g}}{208.2} = 0.015\text{mol}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process.

$$(26.01 - 22.24) - 3.18 = 0.59\text{g of water lost}$$

- b) Calculate the moles of water lost in the heating process

$$n = \frac{m}{M} \quad n(\text{H}_2\text{O}) = \frac{0.59}{18.016} = 0.033\text{moles of water lost}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	BaCl_2	H_2O
mass	3.18g	0.59g
mole	$n = \frac{3.18}{208.2} = 0.015$	$n = \frac{0.59}{18.016} = 0.033 \text{ mol}$
mole rat	$\frac{0.015}{0.015} = 1$	$\frac{0.033}{0.015} = 2.2$
empirical formula $\rightarrow \text{BaCl}_2 \cdot 2\text{H}_2\text{O}$		

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

It implies that some of the water was still combined to the BaCl_2 so when it was heated again more left.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Because the desiccator absorbs the vapour so that it doesn't join with the substance again.

TOTAL = 15 / 15

100%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	21.59g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	29.28g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	28.66g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	28.67g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

Mr(BaCl_2) = $1 \times 137.3 + 2 \times 35.45 = 208.2$

- b) Calculate the moles of anhydrous BaCl_2

$\frac{208.2}{6.022 \times 10^{23}} = 3.45 \times 10^{-24}$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

4.09g

- b) Calculate the moles of water lost in the heating process

~~4.09~~

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

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Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

~~That after the second~~ that all of the H_2O wasn't evaporated

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Because the desiccator cools it alot faster

TOTAL = 3 / 15

20%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	22.24g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	26.01g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	25.42g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	25.42g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$25.42 - 22.24\text{g} = 3.18\text{g}$$

- b) Calculate the moles of anhydrous BaCl_2 $\frac{137.3 + 35.45 + 35.45}{172.15}$ (BaCl_2)

$$n = \frac{m}{M} = \frac{25.42}{172.15} = \frac{3.18}{208.2}$$

$$n = 0.14$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$26.01 - 25.42 = 0.59\text{g}$$

- b) Calculate the moles of water lost in the heating process

$$\text{Water lost} = 0.59$$

$$n = \frac{m}{M} = \frac{0.59}{1.008 \times 2 + 16.00} = \frac{0.59}{33.008} = 0.0178\text{ mol}$$

$$= 0.018\text{ mol}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	Barium chloride	H_2O
Mass	3.18g	0.59g
Molar mass	172.75	18.06
Mass ratio	$\frac{3.18}{172.75} = 0.018$	$\frac{0.59}{18.06} = 0.032$

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

It weighed more because the H_2O hadn't been evaporated!
or heated yet.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

TOTAL = 8 / 15

53%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	25.18
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	29.40
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	28.77
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	28.77

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$m(\text{BaCl}_2) = 28.77 - 25.18 = 3.59 \text{ g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$\begin{aligned} n(\text{BaCl}_2) &= m/M \\ &= 3.59 / 208.2 \\ &= 0.0172 \text{ mol} \end{aligned}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$m(\text{H}_2\text{O}) = 29.40 - 28.77 = 0.63 \text{ g}$$

- b) Calculate the moles of water lost in the heating process

$$\begin{aligned} n(\text{H}_2\text{O}) &= m/M \\ &= 0.63 / 18.016 \\ &= 0.0349 \text{ mol} \end{aligned}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	BaCl_2	H_2O	
mass	3.59 g	0.63 g	
moles	0.0172	0.0361	
Mole ratio	1	2.02	(3)

Empirical formula = $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

chemicals would of reacted, causing a gas which reduces the mass. X

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Other chemicals won't react with it.
air moisture.

TOTAL = 11.5 / 15

77%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	29.69 29.16
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	29.16 29.69
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	29.15
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	29.15

26.16g

①

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$\begin{aligned} & \text{29.16} \quad \text{29.69g} - \text{29.16g} = \text{0.53g} - \text{0.01g} = \text{0.52g} \end{aligned}$$

- b) Calculate the moles of anhydrous BaCl_2

$$\begin{aligned} n &= \frac{0.52\text{g}}{208.2} = 0.002492598 \text{ mol} \\ &= 0.00249 \text{ mol} \end{aligned}$$

③

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$29.16\text{g} - 29.15\text{g} = 0.01\text{g}$$

- b) Calculate the moles of water lost in the heating process

$$\begin{aligned} n &= \frac{0.01\text{g}}{18.016} = 0.000555062 \text{ mol} \\ &= 0.000555 \text{ mol} \end{aligned}$$

③

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

Handwritten working for Question 4:

$\text{H}_2\text{O} = \frac{0.000555 \text{ mol}}{0.002497598 \text{ mol}} = 0.222238326$

~~$\text{BaCl}_2 \cdot \text{H}_2\text{O}$~~

0.052 g (Mass)
 0.51 g (BaCl₂)
 0.01 g (H₂O)

~~$2\text{BaCl}_2 \cdot 19\text{H}_2\text{O}$~~

Answer

Question 5

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

Handwritten answer for Question 5:

This implies that more H_2O was lost and then a little was gained 4.587 (lost) 4.533 & gained 4.535

Question 6

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Handwritten answer for Question 6:

① Its hot so cooling on a laboratory bench is not practical ② Improve validity of experiment

TOTAL = 7.5 / 15

50%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	24.02g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	27.28g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	26.79g
* Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	26.8g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$26.8 - 24.02 = 2.78g \quad (\text{used measurement \#2})$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M}$$

$$n(\text{BaCl}_2) = \frac{2.78}{208.2} = 0.0133 \text{ mol}$$

Question 3

(4 marks)

- * a) Calculate the mass of water lost in the heating process

$$27.28 - 26.8 = 0.48g \quad (\text{BaCl}_2 \cdot x\text{H}_2\text{O})$$

- b) Calculate the moles of water lost in the heating process

$$n = \frac{m}{M} \quad n(x\text{H}_2\text{O}) = \frac{0.48}{18.016} = 0.0266 \text{ mol}$$

4

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	BaCl_2	$x\text{H}_2\text{O}$
mass	2.06 g	0.66 g
$n = \frac{m}{M}$ mole	0.012 mol	0.037 mol
ratio	1	3.08
		≈ 3
empirical formula:	$\text{BaCl}_2 \cdot 3\text{H}_2\text{O}$	

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

The amount of H_2O was not completely gone, and the other two measurements were closer together, implying that the H_2O in $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ had completely gone. The BaCl_2 was pure in the last 2 measurements.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

It ensures the BaCl_2 is at a constant temperature, as well as the fact that the measurements may change when they are at different temperatures.

TOTAL = 13 / 15

87%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	26.10 g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	30.44 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	29.8 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	29.8 g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$m = 3.68 \text{ g} \quad 30.44 - 26.10 = 3.68 \text{ g}$$

$$29.8 - 26.10 = 3.70 \text{ g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} \quad n(\text{BaCl}_2) = \frac{3.68}{208.2} = 0.018 \text{ mol}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process ?

$$\text{amount of water lost} = 0.66 \text{ g} \quad 30.44 - 29.78 = 0.66 \text{ g}$$

- b) Calculate the moles of water lost in the heating process

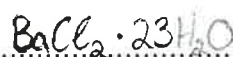
$$n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{0.66}{18.016} = 0.036 \text{ mol}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	BaCl_2	H_2O
mass	4.34	0.66
n.p.s	$\frac{4.34}{208.2} = 0.0016$	$\frac{0.66}{18.016} = 0.037$
ratio	$\frac{0.0016}{0.0016} = 1$	$\frac{0.037}{0.0016} = 23$



You already calculated this...

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

This implies that more water (H_2O) was being burnt off each time the (BaCl_2) was being heated. (evaporated)

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

This allows the BaCl_2 to cool in an enclosed space where there is less oxygen... so the chemicals don't react with as much oxygen to differ the reaction rate and type of reaction. H_2O , Air moisture.

TOTAL = 11 / 15

73%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	25.03
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	28.65
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	28.12
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	28.13

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$28.12 - 25.03 = 3.09 \text{ g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$3.09 / (137.3 + (2 \times 35.45)) = n$$

$$= 3.09 / 208.2$$

$$n = 0.0148 \text{ mol}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$28.65 - 28.12 = 0.53 \text{ g}$$

- b) Calculate the moles of water lost in the heating process

$$0.53 / ((2 \times 1.008) + 16) = n$$

$$= 0.53 / 18.016$$

$$n = 0.0294$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

0.0294 mol of water (H_2O)

~~0.0148~~ 0.0148 mol of barium chloride (BaCl_2)

$$0.0294 \div 0.0148 = 1.986$$

$$= 2$$



Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

When it was heated, the water inside the $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ was turned into water vapour and left the chemical leaving just BaCl_2 . The process ~~was~~ was repeated again for measurement 2 until all water was removed.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

The desiccator will help absorb the water vapour, if ~~the~~ left on a bench, the BaCl_2 might absorb some of the water vapour which will affect results.

TOTAL = 15 / 15

100%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	26.10 g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	30.44 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	29.78 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	29.8 g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$29.78 - 26.10 \text{ g} = 3.68 \text{ g}$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} \quad n(\text{anhydrous } \text{BaCl}_2) = \frac{3.68}{208.2} = 0.0177 \text{ mol}$$

$$M(\text{BaCl}_2) = 208.2 \text{ g mol}^{-1}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$30.44 - 29.8 = 0.64 \text{ g of water lost in heating process}$$

- b) Calculate the moles of water lost in the heating process

$$n = \frac{m}{M} \quad n(\text{H}_2\text{O lost in heat process}) = \frac{0.64}{18} = 0.0356 \text{ mol}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

	Ba	Cl	H	O	% Ba = $\frac{137.3}{208.2} = 65.9\%$
Mass			0.		% Cl = $\frac{70.9}{208.2} = 34.1\%$
					%

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

The recorded results imply that the BaCl_2 compound lost water due to evaporation in the heating process. and had not finished doing so after 1 lot of heating. (0.5)

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

The anhydrous BaCl_2 is cooled in a desiccator so no extra H_2O molecules can condense back into the compound. ✓

TOTAL = 9.5 / 15

63%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	24.31 g
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	27.95 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	27.40 g
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	27.4 g

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$27.95 \text{ g} - 24.31 \text{ g} = 3.64 \text{ g}$$

 $m(\text{BaCl}_2)$

- b) Calculate the moles of anhydrous BaCl_2

$$\frac{24.31}{22.95}$$

$$= 0.880 \text{ mol}$$

① for including units.

 $M(\text{BaCl}_2)$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$27.95 \text{ g} - 27.40 \text{ g} = 0.55 \text{ g}$$

- b) Calculate the moles of water lost in the heating process

$$\frac{0.55}{18.016} = 0.031 \text{ mol}$$

④

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

13%	
$\frac{13\%}{172.75} =$	—

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

Because the mass of Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ vary, due to them being heated constantly & also the Anhydrous $\text{BaCl}_2 \cdot \text{H}_2\text{O}$ was being heated.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Because of the crystallization, and also because the anhydrous BaCl_2 was extremely hot & could melt the science bench.

TOTAL = 7 / 15

47%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	23.5
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	26.8
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	26.29
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	26.29

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$\cancel{26.8 - 23.5 = 3.3 \text{ g}} \quad 26.29 - 23.5 = 2.79$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} = \frac{2.79}{M(137.3 + (2 \times 35.45))} = \frac{2.79}{108.2} = 0.0134 \text{ mol}$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$\cancel{26.8 - 26.29 = 0.51 \text{ g}} \quad 26.8 - 26.29 = 0.51 \text{ g}$$

- b) Calculate the moles of water lost in the heating process

$$n = \frac{m}{M} = \frac{0.51}{M(1.008 \times 2) + 16} = \frac{0.51}{18.016} = 0.0283 \text{ mol}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

3.3g

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

The first recorded mass of the results was different to the next two ~~masses~~ implies that this measurement had more water.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

It is important that the anhydrous BaCl_2 is left to cool in the desiccator because it allows for consistent cooling.

TOTAL = 10.5 / 15

70%

Task 1: Water of Crystallisation of Barium Chloride

Weight: 2%

Complete the experiment using the instructions on the provided sheet. Work in pairs for your experiment but answer questions on this sheet individually. Note that you can answer some questions from the reverse page while your barium chloride is still being heated.

Note: Marks will be deducted if you fail to follow lab procedures or fail to pack away equipment.

Question 1

(2 marks)

Show the results from your experiment in the form of a table.

Object	Mass (g)
Crucible + Lid	25.18
Crucible + Lid + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	29.40
Crucible + Lid + Anhydrous BaCl_2 (Measurement #1)	28.77
Crucible + Lid + Anhydrous BaCl_2 (Measurement #2)	28.77

I didn't get these measurements

Question 2

(4 marks)

- a) Calculate the mass of anhydrous BaCl_2 (anhydrous = without water)

$$28.77 - 25.18 = 3.59$$

- b) Calculate the moles of anhydrous BaCl_2

$$n = \frac{m}{M} \rightarrow \text{Ba} = 137.3 \quad \text{Cl} = 35.45 \times 2 = 70.9$$

$$137.3 + 70.9 = 208.2$$

$$208.2 \div 3.59 = 0.5793$$

Question 3

(4 marks)

- a) Calculate the mass of water lost in the heating process

$$M = 29.40 - 28.77 = 0.63$$

- b) Calculate the moles of water lost in the heating process

$$n = \frac{m}{M} \rightarrow \frac{0.63}{18.016} = 0.035$$

$$n = \frac{m}{M} = \frac{0.63}{18.016} = \text{--- mol}$$

Question 4

(3 marks)

Using your answers to Question 3, calculate the empirical formula of the hydrated barium chloride. Show all of your working and write the formula of the compound in the form of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

mass	BaCl_2	H_2O	
moles	0.017 $\times 0.01$	0.625 $\times 0.01$	
molar ratio	57.93 $\times 1.6$	16 $\times 1.6$	$= 1.856$
	92.688		

Question 5

(1 mark)

The table below shows results from another group.

Object	Mass (g)
Crucible	2.670
Crucible + Hydrated $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$	4.621
Crucible + Anhydrous BaCl_2 (Measurement #1)	4.587
Crucible + Anhydrous BaCl_2 (Measurement #2)	4.533
Crucible + Anhydrous BaCl_2 (Measurement #3)	4.535

The first recorded mass of "Crucible + Anhydrous $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ " in the results table was different to the next two recorded masses. What does this imply?

The weighing scale was faulty.

Question 6

(1 mark)

Why is it important that the anhydrous BaCl_2 is left to cool in a desiccator as opposed to cooling on a laboratory bench?

Because it could react with moisture in the air changing the mass and ruining the experiment.

TOTAL = 7 / 15

47%